JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY ANANTAPUR
COLLEGE OF ENGINEERING (AUTONOMOUS)ANANTHAPURAMU– 515 002 (A.P) INDIA

B. Tech. – I Year I Semester
(Common to EEE, ECE, CSE)

Subject code	Title of the Subject	L	T	P	C
23A10302	Engineering Workshop	0	0	3	1.5

Course Objectives: To familiarize students with wood working, sheet metal operations, fitting and Electrical house wiring skills

Course Outcomes:

- **CO1:** Identify works hop tools and their operational capabilities.
- **CO2**: Practice on manufacturing of components using works hop trades including fitting, carpentry, foundry and welding.
- **CO3:** Apply fitting operations in various applications.
- **CO4**: Apply basic electrical engineering knowledge for House Wiring Practice.
- **CO5:** Demonstration and Practice of plumbing and welding.

SYLLABUS

1. **Demonstration**: Safety practices and precautions to be observed in workshop.
2. **Wood Working:** Familiarity with different types of wood sand tools used in wood working and make following joints.
   a) Half–Lap joint b) Mortise and Ten on joint c) Corner Dovetail joint or bridle joint d) Demonstration of Power tools
3. **Sheet Metal Working**: Familiarity with different types of tools used in sheet metal working, Developments of following sheet metal job from GI sheets.
   a) Tapered tray b) Conical funnel c) Elbow pipe d) Brazing
4. **Fitting:** Familiarity with different types of tools used in fitting and do the following fitting exercises.
   a) V-fit b) Dovetail fit c) Semi-circular fit d) Bicycle tire puncture and change of two-wheeler tire
5. **Electrical Wiring**: Familiarity with different types of basic electrical circuits and make the following connections.
   a) Parallel and series b) Two-way switch c) Go down lighting
   b) d)Tube light e) Three phase motor f) Soldering of wires
6. **Foundry Trade:** Demonstration and practice on Molding tools and processes, Preparation of Green Sand Molds forgiven Patterns.
7. **Welding Shop**: Demonstration and practice on Arc Welding and Gas welding. Preparation of Lap joint and Butt joint.
8. **Plumbing:** Demonstration and practice of Plumbing tools, Preparation of Pipe joints with coupling for same diameter and with reducer for different diameter

Textbooks:

1. Basic Workshop Technology: Manufacturing Process, Felix W.; Independently Published, 2019.Workshop Processes, Practices and Materials; Bruce J. Black, Routledge publishers, 5th Edn.2015.
2. A Course in Workshop Technology Vol I. & II, B.S. Raghuwanshi, Dhanpath Rai & Co., 2015 &2017.

Reference Books:

1. Elements of Workshop Technology, Vol. I by S.K. Hajra Choudhury &Others, Media Promoters and Publishers, Mumbai.2007,14thedition
2. Workshop Practice by H.S. Bawa, Tata-McGrawHill,2004.
3. Wiring Estimating, Costing and Contracting; Soni P.M. &Upadhyay P.A.; Atul Prakashan, 2021-22.

Mapping between Course Outcomes and Programme Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1					✓	✓	✓		✓			
CO2					✓						✓	✓
CO3						✓					✓	✓
CO4						✓			✓		✓	✓
CO5						✓			✓		✓	✓